

Nguyen Minh Khoa

AI Engineer & Researcher

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I'm Nguyen Minh Khoa, an AI engineer and Research Assistant at AiTA Lab, FPT University.

My recent work includes handwritten mathematical expression recognition and text-to-molecule generation. I work across data preparation, model experiments, and inference integration, with a preference for research ideas that can be tested in working software.

My current focus is AI for intelligent manufacturing. I look for work that gives equal weight to careful experiments and hands-on implementation.

I'm open to AI engineering, research engineering, and applied machine-learning collaborations.

RESEARCH EXPERIENCE

Research Assistant | AiTA Lab, FPT University | Oct 2024 - Present

- Co-authored two HMER papers at ICDAR and in the Journal of Information and Telecommunication, contributing masked-language pretraining and multi-task learning experiments.
- First-authored LexiChem (APWeb-WAIM 2026): developed methods, ran ablations and presented the work in Da Nang. Co-first author of ChemAligner-T5 (ICCIES 2026).
- Co-authored CorrTie (APWeb-WAIM 2026), contributing critical review and methodological refinement.

SELECTED PROJECTS

LexiChem | AI & Inference Contributor | 2025 - 2026

- Built the AI inference workflow so users can generate molecules from text, validate structures and revisit saved runs using FastAPI, Triton, RDKit and MongoDB.
- Built a three-dataset training pipeline and ran ablations for the first-author APWeb-WAIM paper.

NexOps | AI, Simulation & Platform Lead | 2025 - Present

- Built the local MES/SCADA investment/demo prototype connecting simulated machines to production planning, alarms and OEE dashboards through MQTT, FastAPI and TimescaleDB.
- Connected a Bambu Lab printer over LAN, demonstrating real device control in the local prototype.

SELECTED PUBLICATIONS

LexiChem: Shared-Latent Alignment for Faithful Text-to-Molecule Generation in SELFIES Space

APWeb-WAIM 2026 | First author | Accepted & presented

Mask CoMER: Enhancing Handwritten Mathematical Expression Recognition with Masked Language Pretraining and Regularization

ICDAR 2025 | Co-author | Published

Enhancing handwritten mathematical expression recognition with convolutional block attention refinement module and multi-task learning

Journal of Information and Telecommunication | Co-author | Published online 19 August 2026

EDUCATION

Bachelor of Science in Artificial Intelligence | FPT University, HCMC

2022 - 2026 | Academic requirements completed; degree conferral expected November 2026.

AWARDS & SCHOLARSHIPS

50% tuition scholarship, FPT University (2022) | **RESFES Second Prize**, 2025 & 2026.

TrendRadar team awards/funding: Demo Pitching Day, VND 50M; Innovation Quest, VND 40M (2025).

TECHNICAL SKILLS

AI & research: Python, PyTorch, Transformers, model experiments and ablations, HMER, cross-modal learning, text-to-molecule generation

Systems & data: FastAPI, NVIDIA Triton, RDKit, PostgreSQL/TimescaleDB, MongoDB, MQTT, Docker, Git

SELECTED CERTIFICATIONS

AIO | AI VIET NAM: Machine Learning; Deep Learning; Computer Vision & NLP; GenAI & LLMs.

Machine Learning (Stanford Online / DeepLearning.AI, 2023) | Deep Learning (DeepLearning.AI, 2024)